

EconoPulse AI: Autonomous Multi-Modal Global Intelligence & Geospatial Digital-Twin Platform

Live Threat Telemetry, 3D Digital-Twin Globe, Zero-Fabrication OpenStreetMap Spatial Search & Persona Impact Synthesis

PROJECT TITLE:	EconoPulse AI — Real-Time Global Intelligence & Geospatial Telemetry Platform
PARTICIPANT NAME:	Sumanth Duraimurugan
REGISTER / ROLL NO:	25cu0310160
INSTITUTION:	Hindustan Institute of Science and Technology (HITS)
TEAM STRUCTURE:	Individual Participant (Solo Entry)

PROJECT ABSTRACT

Exact Count: 185 words (within 150–200 limit)

Global crises, supply disruptions, and regional emergencies unfold rapidly, yet decision-makers and citizens struggle with delayed reporting, fragmented data streams, and lack of localized situational awareness. The aim of this project is to develop EconoPulse AI, an autonomous global intelligence and geospatial digital-twin platform that synthesizes live macro-level shocks with real-world local infrastructure discovery. Our proposed solution unifies real-time event telemetry with an interactive 3D Three.js digital-twin globe and a synchronized OpenStreetMap street-level viewer, enabling users to seamlessly transition from planetary threat monitoring to street-level inspection. Built with React, Node.js, Express, Three.js, Overpass API, Nominatim geocoding, and Google Gemini LLMs, the platform integrates category-coded spatial telemetry—featuring luminous red markers for healthcare facilities and distinct signatures for transit, academic, and civic centers—coupled with adaptive altitude zooming across global, state, and neighborhood scales. What makes this idea innovative is its zero-fabrication architecture, fusing verified geospatial data with AI-driven persona impact projections and voice navigation to translate macro crises into tailored personal guidance. The expected outcome is a unified intelligence dashboard that empowers users with rapid emergency navigation, proactive risk mitigation, and actionable geospatial awareness during worldwide disruptions.

ENGINEERED SYSTEM CAPABILITIES & INNOVATION MATRIX

FEATURE MODULE	IMPLEMENTATION & DATA PIPELINE	USER & OPERATIONAL BENEFIT
3D Digital-Twin Globe	Three.js planetary sphere, smooth OrbitControls, and GPU-accelerated spatial coordinate projection.	Fluid worldwide exploration and spatial context of emerging global disruptions.
Adaptive Regional Zoom	Dynamic altitude scaling navigating across Global (275), Regional/State (145), and Local (120) views.	Instant zoom into exact geographic regions matching contextual search queries.

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FEATURE MODULE	IMPLEMENTATION & DATA PIPELINE	USER & OPERATIONAL BENEFIT
Visual Category Telemetry	Luminous red markers (#ef4444) for hospitals/healthcare, blue for colleges/schools, amber for transit.	Immediate visual triage and rapid emergency facility identification.
Zero-Fabrication OSM Search	Real-world Overpass API querying and Nominatim geocoding; strictly rejects synthetic coordinate hallucination.	100% trustworthy, ground-truth physical locations with verified GPS coordinates.
AI Persona Impact Analysis	Google Gemini LLM reasoning projecting macro financial, logistical, and personal operational risks.	Translates abstract global events into actionable, role-tailored guidance.

Candidate: Sumanth Duraimurugan • Roll: 25cu0310160 • Hindustan Institute of Science and Technology
Live Prototype: <https://global-intel-telemetry.vercel.app>